

Chapter - 13 Biodiversity and Conservation

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"Biodiversity is the variety of life on Earth"

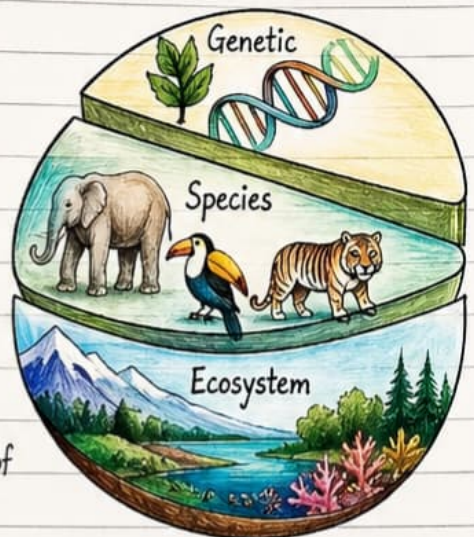
1. BIODIVERSITY

It is the totality of genes, species and ecosystems of a region.

- India is one of the 12 megadiverse countries of the world.

Levels of Biodiversity

- ① **Genetic Diversity**: Variation in genes within a species.
- ② **Species Diversity**: Variety and richness of species in an area.
- ③ **Ecosystem Diversity**: Variety of ecosystems in a region.



2. PATTERNS OF BIODIVERSITY

- Biodiversity is not uniform; it varies from place to place.
- Two important patterns:

(i) Latitudinal Gradient

Species richness increases from poles towards equator.
(Tropics > Temperate > Polar)

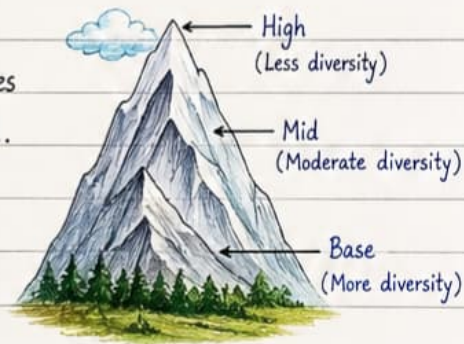


Example: Amazon rainforest (Tropical region) has highest diversity.

(ii) Altitudinal Gradient

Species richness decreases with increase in altitude.
(Base > Mid-altitude > High altitude)

Example: Himalayas show very less diversity at high altitudes.



3. IMPORTANCE OF BIODIVERSITY

- ① **Ecological balance**: Maintains stability of ecosystems.
- ② **Economic value**: Provides food, medicine, timber, fibre, fuel, etc.
- ③ **Ethical value**: Every species has a right to exist.
- ④ **Social & cultural value**: Inspired art, culture, religion and traditions.
- ⑤ **Scientific value**: Useful for research and discovery of new species and drugs.



Did You Know?

India has about 8% of the world's recorded species but only 2.4% of the world's land area.

Quick Revision

- ★ Biodiversity = Genes + Species + Ecosystems
- ★ Not uniform → changes with latitude & altitude
- ★ Essential for ecological balance & human welfare.



4. LOSS OF BIODIVERSITY

Biodiversity is lost due to two major reasons:

(i) Natural Causes

- Climatic changes (e.g., drought, flood, temperature extremes)
- Natural disasters (e.g., earthquakes, volcanoes, forest fires)
- Evolutionary processes

(ii) Anthropogenic Causes

- Habitat loss & fragmentation (deforestation, urbanization, mining)
- Over exploitation (hunting, poaching, overfishing)
- Invasive alien species
- Pollution (air, water, soil)
- Climate change (global warming)

Human activities are the major cause.

IUCN RED LIST CATEGORIES

IUCN (International Union for Conservation of Nature) assesses the risk of extinction of species.

- EX** Extinct: No reasonable doubt that the last individual has died.
- EW** Extinct in the Wild: Survive only in cultivation or in captivity.
- CR** Critically Endangered: Extremely high risk of extinction.
- EN** Endangered: High risk of extinction in the wild.
- VU** Vulnerable: High risk of endangerment in the wild.
- NT** Near Threatened: Likely to qualify for a threatened category in near future.
- LC** Least Concern: Widespread and abundant.

5. CONSERVATION OF BIODIVERSITY

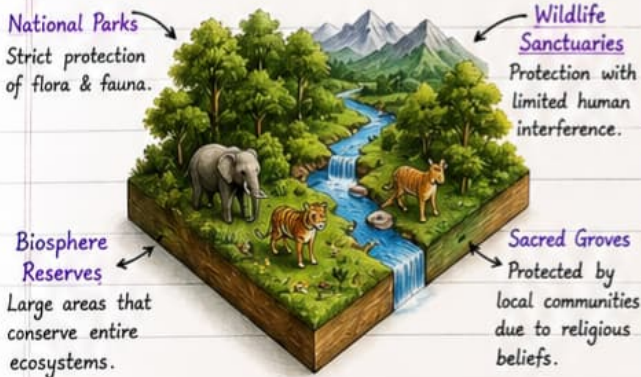
Conservation means protection, maintenance and wise use of biodiversity.

(i) In-situ Conservation.

Conservation of species in their natural habitats.

Examples:

- National Parks
- Wildlife Sanctuaries
- Biosphere Reserves
- Sacred Groves

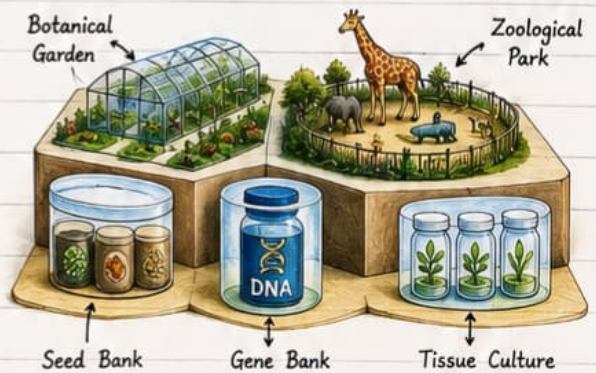


(ii) Ex-situ Conservation

Conservation of species outside their natural habitats under controlled conditions.

Examples:

- Botanical gardens
- Zoological parks
- Seed banks
- Gene banks
- Tissue culture



6. BIODIVERSITY HOTSPOTS OF INDIA

India is one of the 17 mega-diverse countries. It has 4 biodiversity hotspots.

Hotspot	Major Features
Western Ghats	Rich in endemic plants, amphibians, reptiles.
Himalaya	Diverse altitude, many endemic species.
Indo-Burma	High species richness, tropical forests.
Sundaland (Nicobar Islands)	Unique flora & fauna, coral reefs, mangroves.

Remember!

Save biodiversity
→ Save life.

Every species has a role in nature's balance.

Key Points

- ★ Biodiversity is our life support system.
- ★ Loss of biodiversity is a threat to future generations.
- ★ Conservation is the need of the hour.
- ★ Protect nature, because we are a part of it.



7. CONSERVATION STRATEGIES

Two main approaches are used to conserve biodiversity.

(A) In-situ Conservation (On-site)

- It involves conservation of species within their natural habitats.

Examples :

- National Parks
- Wildlife Sanctuaries
- Biosphere Reserves
- Sacred Groves

Protection of whole ecosystem and natural processes.

Maintains ecological balance and genetic diversity.

Allows species to evolve naturally.

More effective for long term conservation.



(B) Ex-situ Conservation (Off-site)

- It involves conservation of species outside their natural habitats under controlled conditions.

Examples :

- Botanical Gardens
- Zoological Parks
- Seed Banks
- Gene Banks
- Tissue Culture



Botanical Garden



Zoological Park



Seed Bank



Gene Bank



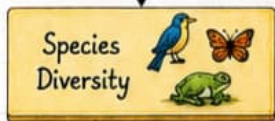
Tissue Culture

Useful for endangered species, research, breeding and reintroduction programmes.

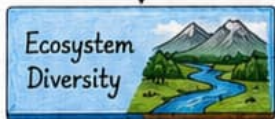
8. BIODIVERSITY AT DIFFERENT LEVELS



Variation in genes within a species.
Example : Different varieties of rice, mango, humans.



Variety and richness of species in a given area.
Example : Different animals in a forest.



Variety of ecosystems in a region.
Example : Forests, deserts, wetlands, grasslands, oceans.

9. THREATS TO BIODIVERSITY

- **Habitat Loss and Fragmentation :** Destruction of forests, wetlands, grasslands due to human activities.
- **Overexploitation :** Excessive hunting, fishing, timber cutting and collection of medicinal plants.
- **Invasive Alien Species :** Introduced species outcompete native species and disturb ecosystems.
- **Pollution :** Air, water and soil pollution harm organisms and their habitats.
- **Climate Change :** Global warming and extreme weather events affect species survival.
- **Co-extinction :** Loss of one species leads to loss of dependent species.

10. SUSTAINABLE USE OF BIODIVERSITY

Biodiversity should be used wisely so that it remains available for future generations.

- Use resources within their regenerative capacity.
- Reduce, Reuse and Recycle resources.
- Promote organic farming and eco-friendly practices.
- Conserve forests, water and wildlife.
- Protect traditional knowledge of local communities.

Think Green!



Quick Revision

- ★ Biodiversity = Variety of life
- ★ India is one of the 17 mega-diverse countries.
- ★ Conservation = Protection + Wise use
- ★ In-situ = On-site (Natural habitats)
- ★ Ex-situ = Off-site (Controlled conditions)
- ★ Save biodiversity → Save Earth



The future of humanity depends on the health of our ecosystems.
Let's conserve today for a better tomorrow.



11. CONSERVATION IN INDIA

India has a long tradition of conserving nature. Many protected areas have been established to conserve biodiversity.

(A) National Parks

- Large areas set aside to protect wild animals and their habitats.
- No human activity is allowed, except tourism and research.

Example :

Jim Corbett NP (Uttarakhand)

Kaziranga NP (Assam)



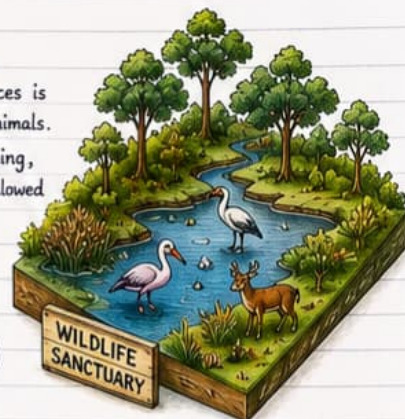
(B) Wildlife Sanctuaries

- Areas where use of resources is regulated to protect wild animals.
- Human activities like grazing, collection of firewood are allowed in limited manner.

Example :

Periyar WS (Kerala)

Bharatpur WS (Rajasthan)



(C) Biosphere Reserves

- Large natural areas for conservation, research, education and sustainable use.
- It has three zones :

Example :

Nilgiri Biosphere Reserve (TN, KL, KA)

Nanda Devi Biosphere Reserve (Uttarakhand)



Core Zone

Strict protection.
No human activity.

Buffer Zone

Limited human activities
like research, education.

Transition Zone

Area of settlements,
farming and other
sustainable activities.

12. SOME IMPORTANT INDIAN INITIATIVES

Initiative / Act	Key Features
Project Tiger (1973)	- Launched to save tigers and their habitats. - It helped in increasing tiger population.
Project Elephant (1992)	- Aims at the conservation of elephants and their habitats. - Focus on reducing human-elephant conflict.
Sacred Groves	Patches of forest conserved by local communities due to religious beliefs.
Biodiversity Act (2002)	Provides legal framework for conservation, sustainable use and fair sharing of benefits.
Chipko Movement	People (mainly villagers, especially women) hugged trees to stop their cutting in the Himalayas.

Remember !

- Conservation is not just about saving animals or plants, but about maintaining ecological balance.
- Every species, no matter how small, has a role in the ecosystem.
- Loss of even one species can disturb the whole chain of life.
- Protect biodiversity today for a sustainable tomorrow.



13. BIODIVERSITY HOTSPOTS IN INDIA

India has four biodiversity hotspots because of its high species richness and endemism.

Endemic Species

Species that are found naturally in a particular geographical area and nowhere else.

Hotspot	Regions Covered	Major Features	Example of Endemic Species
Western Ghats	Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu	Rich in amphibians, reptiles, birds and flowering plants.	Lion-tailed Macaque, Malabar Civet, Nilgiri Tahr
Himalaya	Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh	High altitude, diverse forests, medicinal plants.	Snow Leopard, Himalayan Monal, Red Panda
Indo-Burma	Arunachal Pradesh, Nagaland, Manipur, Mizoram, etc.	Tropical forests, very high species richness.	Hoolock Gibbon, Hornbill, Burmese Python
Sundaland (Nicobar Islands)	Nicobar Islands (Andaman & Nicobar Islands)	Coral reefs, mangroves, unique marine life.	Nicobar Megapode, Giant Robber Crab, Nicobar Pigeon

Final Thought

Conserve biodiversity → Conserve life → Conserve our future.

